
Tutorial 5

Malware Threats

Computer Security BCS2420

Part A

1. Which statement about the *undecidability* of detecting malware is correct?
 - ☐ It is theoretically proven that a perfect malware detector cannot exist in all cases.
 - ☐ Malware detection is easily automated with zero false positives.
 - ☐ Any code that modifies system files is necessarily malware.
 - ☐ Commercial antivirus tools already achieve perfect detection.
2. Which feature distinguishes a *worm* from a *virus*?
 - ☐ Worms require user intervention (e.g., opening an attachment), while viruses do not.
 - ☐ Worms spread autonomously through network vulnerabilities, whereas viruses typically rely on users executing infected files.
 - ☐ Worms cannot replicate, but viruses replicate aggressively.
 - ☐ Worms are always detected by signature-based antivirus, while viruses often go undetected.
3. A piece of malware that intercepts system calls to hide its processes from user utilities is typically classified as:
 - ☐ A logic bomb
 - ☐ A worm
 - ☐ A rootkit
 - ☐ Ransomware

4. Which best describes a *ransomware* attack?
- ☐ Malware that primarily spreads via self-replicating network exploits
 - ☐ A Trojan horse that silently steals keystrokes for indefinite periods
 - ☐ Malware that encrypts user data and demands payment to restore access
 - ☐ A rootkit that replaces the kernel's system-call table
5. When malware uses *polymorphic* techniques, it means that:
- ☐ The malware's encrypted body changes with each infection to evade signature detection.
 - ☐ The malware cannot be disinfected without reformatting the system.
 - ☐ It specifically targets multiple operating systems equally.
 - ☐ The malware does not replicate but changes user privileges frequently.
6. Which factor is *most* associated with a *drive-by download* attack?
- ☐ Relying on user curiosity to click an email attachment
 - ☐ Exploiting browser or plugin vulnerabilities without user awareness
 - ☐ Attaching malicious macros to Microsoft Office documents
 - ☐ Replacing user-mode libraries after the OS boots
7. What is the main purpose of hooking or replacing system calls in a rootkit?
- ☐ To detect other malicious software by scanning the kernel
 - ☐ To stealthily manipulate API results, hiding the rootkit's presence or actions
 - ☐ To accelerate cryptographic operations
 - ☐ To forcibly reinstall the operating system in a loop
8. Which of the following statements about *botnets* is **false**?
- ☐ Botnets typically consist of compromised machines under remote command-and-control.
 - ☐ Botnets may communicate using peer-to-peer or multi-tiered architectures.
 - ☐ A single centralized server is the only way to manage

botnets.

- ☐ Botnets can be used for DDoS attacks, spam campaigns, or credential theft.

9. A *keylogger* is primarily designed to:

- ☐ Render the OS unbootable until a ransom is paid
- ☐ Replace standard binaries (like `ls` or `ps`) to hide malicious processes
- ☐ Capture user keystrokes to harvest passwords or other sensitive data
- ☐ Create duplicates of infected files to spread to other machines

10. Which technique might a kernel-mode rootkit use to remain undetected?

- ☐ Including a special self-destruct button in the user interface
 - ☐ Manipulating the kernel's internal data structures (DKOM) to hide processes or network connections
 - ☐ Encouraging the user to run system updates frequently
 - ☐ Conducting a phone-based phishing campaign
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Part B

1. **Malware Installation Pathways**

- **Question:** Summarize **three** different malware delivery methods (e.g., email attachments, drive-by downloads) and describe **one** real-world example for *each* method.

2. **Virus vs. Worm Differences**

- **Question:** Compare and contrast viruses and worms in terms of **infection mechanism**, **propagation**, and **user involvement**. What implications do these differences have for designing effective defensive measures?

3. **Rootkit Detection Challenges**

- **Question:** Why are rootkits considered especially difficult to detect and remove? Discuss **two** stealth techniques rootkits commonly use and how these pose challenges to traditional antivirus.

4. **Polymorphic and Metamorphic Malware**

- **Question:** Explain the difference between *polymorphic*

and *metamorphic* malware. Illustrate how each makes signature-based detection difficult.

5. **Ransomware Workflows**

- **Question:** Outline the **key steps** in a modern ransomware attack, from initial compromise to payment demand. How do advanced ransomware strains ensure that even offline backups might be compromised?

6. **Botnet Control Structures**

- **Question:** Describe **two** different C2 (command-and-control) architectures for botnets (e.g., centralized vs. peer-to-peer). For each architecture, note a **strength** and a **weakness** from the botmaster's perspective.

7. **Drive-by Download Mechanisms**

- **Question:** Analyze a typical drive-by download scenario. How do attackers exploit **browser vulnerabilities**, and what **redirection** or **obfuscation** techniques might they use?

8. **Kernel-Mode vs. User-Mode Rootkits**

- **Question:** Distinguish between kernel-mode and user-mode rootkits in terms of system privileges, potential stealth, and risk. Why might a kernel-mode rootkit be more powerful?

9. **Trojan Horses and Backdoors**

- **Question:** Many Trojan horses include hidden backdoors. Discuss **how** attackers typically embed these backdoors and **why** they are valuable for long-term system compromise.

10. **Vulnerability Exploits and Malware**

- **Question:** Malware often relies on software vulnerabilities to gain elevated privileges or execute arbitrary code. Describe **two** example vulnerabilities (e.g., buffer overflow, SQL injection) commonly targeted by malware, and why patching them quickly is crucial.
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Part C

1. **Worm Spread Rate**

- **Question:** Suppose a worm scans the internet for vulnerable hosts at a rate of (10^6) probes per hour. If roughly 0.01% of all hosts probed are unpatched and become newly infected, estimate how many **new infections** occur per hour. If the infected population doubles each hour, how many infections might there be after **5 hours**?

2. **Ransomware File-Encryption Scale**

- **Question:** A ransomware variant encrypts user files using a **hybrid** approach. It generates a 256-bit AES key per system to encrypt all files, then encrypts that AES key using a 2048-bit RSA public key. If the **RSA** encryption step takes 20ms on average, how long would it take to encrypt the AES key across **5000** infected systems?

3. **Botnet Attack Throughput**

- **Question:** A botnet with **50,000** active zombies can each send **10,000** spam emails per hour. What is the total spam-sending capacity per day (24 hours)? Express your answer in billions of emails per day.

4. **Polymorphic Malware Signature Coverage**

- **Question:** A polymorphic malware variant can create **100 distinct** decryptor shapes per day. An antivirus vendor can only generate **1 new** signature update per day. Explain *quantitatively* how the malware's mutation rate might outpace signature-based detection (e.g., how many forms remain undetected over a week).
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